

AI-Driven Smart Bio-Digester for Sustainable Waste Management

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ABSTRACT:

This study presents an AI-driven smart bio-digester system designed to convert organic waste into renewable biogas efficiently. The system integrates anaerobic microbial digestion with real-time sensor monitoring to optimize environmental conditions such as temperature, pH, and gas output. By leveraging data analytics and adaptive control mechanisms, the proposed model enhances digestion efficiency and ensures consistent biogas production. The solution addresses critical challenges in waste management by reducing landfill dependency and methane emissions while promoting sustainable energy generation. Additionally, the system supports decentralized deployment, making it suitable for households, hostels, and small communities. The integration of intelligent monitoring not only improves operational performance but also minimizes manual intervention. This approach demonstrates the potential of combining biotechnology with artificial intelligence to create scalable, eco-friendly solutions for modern environmental issues.

KEYWORDS: Bio-digester, Artificial Intelligence, Biogas, Waste Management, Sustainability